COLLEGE OF INNOVATION & TECHNOLOGY

Mid Term Exam

Fall Semester: 2025

Paper Information			
Course Title:	**Algorithm and Complexity Analysis**	**Course Code:**	**CSC-379-01 / CSC-575-01**
Exam Duration:		**Number of Pages:** (including cover page)	**8**
Exam Day/Date:	**10/28/2025**	**Exam Time:**	**10:30 AM**

Student Details			
Student Name:	**Bradley Moore**	**Student ID:**	**35737785**
CRN:	**CSC-575**	**Instructor:**	**Dr. Makram SOUI**
Deadline of submission:	**10/30/2025**	**Time:**	**12:30 PM**

Exam Instructions

- Fill the above 'Student Details' section carefully and complete.
- Cheating is strictly prohibited and action will be taken.

Marking Scheme	
Question No.	**Actual Score / Max. Score**
Q1	**/20**
Q2	**/20**
Q3	**/60**
Total Score:	**/ 100**

Learning Outcomes Achievement		
Learning Outcome	**Question #**	**Student's Achievement**

Q1) Color with red ONE correct answer (letter) for each of the following MCQs in the table below.

1	2	3	4	5	6	7	8	9	10
a	a	a	a	a	a	a	a	a	a
b	b	b	b	b	b	b	b	b	b
c	c	c	c	c	c	c	c	c	c
d	d	d	d	d	d	d	d	d	d

Actual Score
--

1. The performance of an algorithm is most closely related to what?
   a. The total number of element visits
   b. The total number of elements
   c. The type of elements
   d. The number of lines of code in the method

2. While searching in a sorted array which algorithm performs better
   a. Linear Search
   b. Sequential Search
   c. Both Binary and Sequential take same time
   d. Binary Search

3. _____is a good data structure to use for storing things that need to be kept in order, such as a set of documents waiting to be printed on a network printer.
   a. A queue
   b. an array
   c. Tree
   d. Graph

4. _____ consists of a number of nodes, each of which has a reference to the next node.
   a. A linked list.
   b. an array
   c. Tree
   d. Graph

5. Which one of the following statements is a valid declaration of an array named names of three elements?
   a. String names {} = new String[3];
   b. String [] names = new String [3];
   c. String names [3] = new String;
   d. String names[length] = new String();

6. Insertion of an element at the middle of a linked list requires the modification of how many pointers?
   a. 1
   b. 2
   c. 3
   d. 4

7. Process of inserting an element in stack is called _____
   a. Push
   b. Create
   c. Evaluation
   d. Pop

8. Process of removing an element from stack is called _____
   a. Pop
   b. Create
   c. Push
   d. Evaluation
   e.

9. Queue follows _____ which is a form of access to add and remove nodes from a queue.
   a. FIFO, First In First Out.
   b. LIFO, Last In First Out
   c. Both a and b
   d. None of these

10. Which form of access is used to add or remove nodes from a stack?
   a. LIFO, Last In First Out
   b. FIFO, First In First Out
   c. Both A and B
   d. None of these

[*20 Marks*]

Q2) For each question, color True (T) or False (F) as appropriate.

Actual Score
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1	Queue add items to one end of the queue which is called a head	T	F
2.	Stack allows insertion and removal of elements only at one end which is called the top of the stack.	T	F
3	Data structures provide a means of managing large amounts of data, enabling efficient searching, sorting, insertion, and deletion of data.	T	F
4	Computer using the queue for print jobs.	T	F
5	Stacks are different than linked lists or arrays because we can't know the contents of the stack without removing the top element every time	T	F
6	An array is a data structure that contains a group of elements of the same data type.	T	F
7.	A list is an ordered set of data. It is often used to store objects that are to be processed sequentially.	T	F
8.	Data enters the queue at one end and leaves at the other.	T	F
9.	Character is an example of a primitive data structure.	T	F
10.	A linked list is an example of a primitive data structure.	T	F

[*20 Marks*]

Q3)

Actual Score

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A)

Write a java program to obtain the following statements:

1. Create a linkedlist named "Books" with these elements: " Java ", " DB " "Python", and "Network".
2. Add new element "Security" after the element "DB".
3. Display all element of the list.

Note: Please use the in-built Java library class"LinkedList" and interface"ListIterator"

```
////import LinkedList module
 import java.util.LinkedList;
//import ListIterator module
import java.util.ListIterator;

//Create Class
public class QuestionThreeA{
   public static void main(String[] args) {
      //Create Linked List instance call Books
      LinkedList<String> Books = new LinkedList<String>();

      //Add Elements
      Books.add("Java");
      Books.add("DB");
      Books.add("Python");
      Books.add("Network");

      //Add Security after DB
      ListIterator<String> myiterator =  Books.listIterator();
      //move iterator between Java and DB
      myiterator.next();
```

```
    //move iterator between DB and Python
     myiterator.next();
     //Add Security
     myiterator.add("Security");
     //move back to beginning
     myiterator.previous();
     myiterator.previous();
     myiterator.previous();
     //Display all elements
     while(myiterator.hasNext())
     System.out.println(myiterator.next());

  }
}
```

```
PS C:\Users\bradm\OneDrive\Documer
 '-XX:+ShowCodeDetailsInExceptionM
ionThreeA'
Java
DB
Security
Python
Network
```

[*20 Marks*]

B) What is the output of the following program:

```
import java.util.Stack;

public class StackExample {
    public static void main(String[] args) {
        // Create a new stack
        Stack<Integer> stack = new Stack<>();

        // Push elements onto the stack
        stack.push(1);
        stack.push(2);
        stack.push(3);
        stack.push(4);

        // Pop elements from the stack
        while(!stack.isEmpty()) {
            System.out.println(stack.pop());
        }
    }
}
```

Output :

4

3

2

1

Explanation:

Elements are pushed onto the stack. The items are then removed from the stack in LIFO order

[*20 Marks*]

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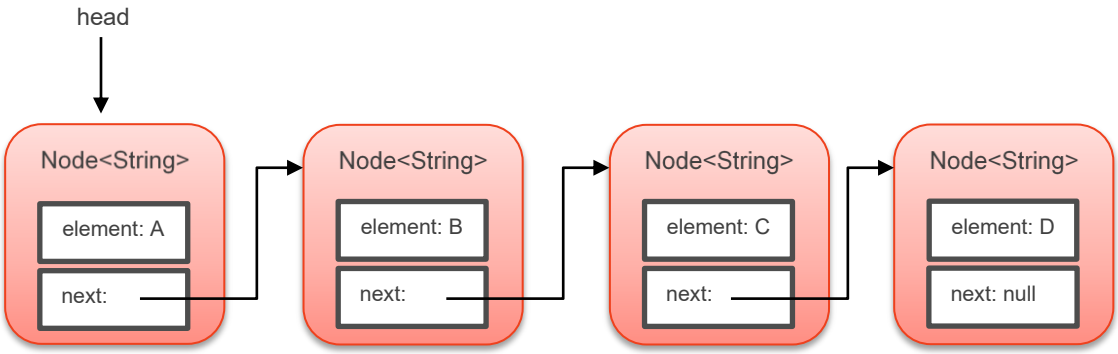
C)

Given the class Node below. We have a linked list with `head` node "A". Write the code to remove the node "C" in the following linked list.

```
public class Node {
     public String element;
     public Node next;
}
```


~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

```
public class LinkedList{
    private Node head;
    private Node tail;
    private int length;

    //would normally run in Main file but for brevity I will run in this file
    public static void main(String[] args) {
        //Create Linked with with initial element
        LinkedList myLinkedList = new LinkedList("A");
        //append elements
        myLinkedList.append("B");
        myLinkedList.append("C");
        myLinkedList.append("D");
        //Remove element C at index 2
        System.out.println(myLinkedList.remove(2).element);

    }

    class Node{
        String element;
        Node next;

        Node(String element){
            this.element = element;
        }
    }

    public LinkedList(String element){
        Node newNode = new Node(element);
        head = newNode;
        tail = newNode;
        length = 1;
    }

    public void append(String element){
        Node newNode = new Node(element);

        if (length == 0){
            head = newNode;
            tail = newNode;
        }else{
            tail.next = newNode;
            tail = newNode;
```

```
        tail.next = null;
    }
    length+=1;
}

public Node removeLast(){
    if (length == 0){
        return null;
    }
    Node temp = head;
    Node pre = head;
    while (temp.next != null){
        pre = temp;
        temp = temp.next;
    }
    tail = pre;
    tail.next = null;
    length--;
    if (length == 0){
        head = null;
        tail = null;
    }
    return temp;
}

public Node removeFirst(){
    if (length == 0){
        return null;
    }
    Node temp = head;
    head = temp.next;
    temp.next = null;
    length--;
    if (length ==0){;
        tail = null;
    }
    return temp;
}

public Node get(int index){
    if (index < 0 || index > length){
        return null;
    }
    Node temp = head;
    for (int i=0; i<index; i++){
        temp = temp.next;
```

```
        }
        return temp;
    }

    public Node remove(int index){
        if (index < 0 || index >= length){
            return null;
        }else if(index == 0){
            return removeFirst();
        }else if (index == length-1){
            return removeLast();
        }else{
            Node prev = get(index-1);
            Node curr = get(index);
            prev.next = curr.next;
            curr.next = null;
            length--;
            return curr;
        }
    }
}
```

```
PS C:\Users\bradm\OneDrive\Documents\GitH
 '-XX:+ShowCodeDetailsInExceptionMessages
dList'
C
PS C:\Users\bradm\OneDrive\Documents\GitH
```

[*20 Marks*]

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End of Examination… Good Luck!